Predicting Loyalty at Turtle Games

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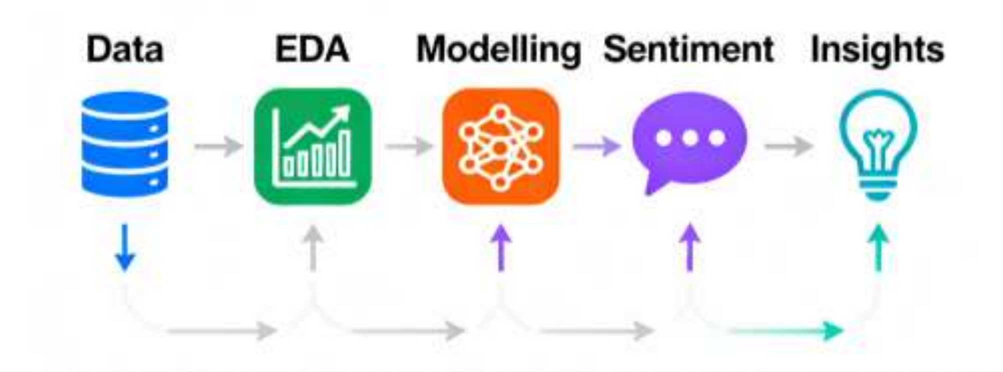
Business Context

- **Objective:** Improve overall sales performance through data-driven insights.
- **Business Questions:**
  - How do customers engage with and accumulate loyalty points?
  - How can customers be segmented for targeted marketing?
  - How can reviewing text data guide marketing improvements?
  - Can loyalty data support predictive modelling?

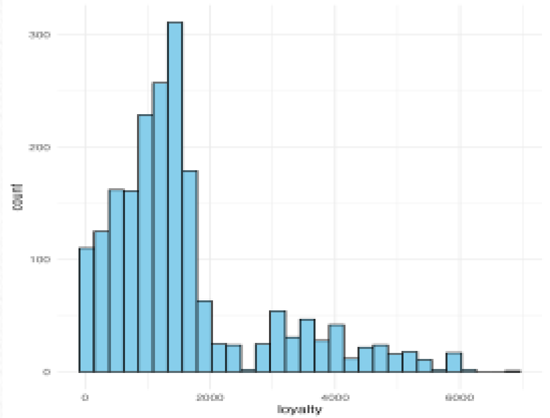
Analytical Framework and Techniques

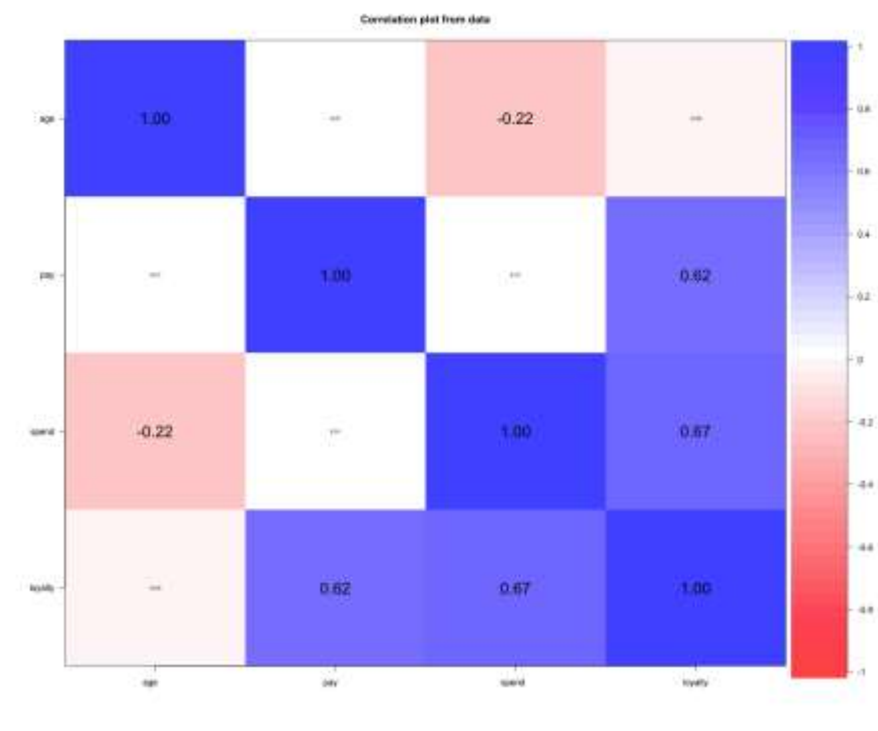
- **Data Source:** turtle_reviews.csv
- **Tools:** Python, R
- **Key Methods:**
    - Data Cleaning & Validation
    - Descriptive Statistics (distribution, skewness, kurtosis)
    - Multiple Linear Regression
    - Decision Tree Regression
    - K-Means Clustering
    - Sentiment Analysis (NLP)



Data Suitability

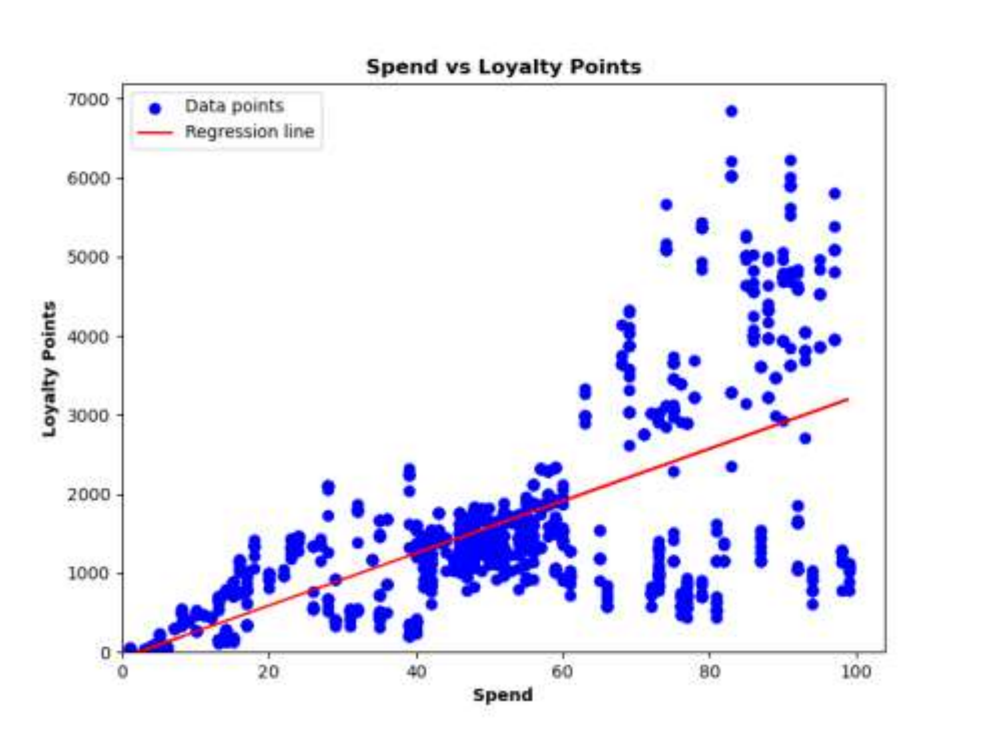
- Loyalty points: **mild right skew**, moderate kurtosis → acceptable for regression after outlier handling
- Correlation highlights: spend–loyalty **r = 0.67**, pay–loyalty **r = 0.62**
- Age–spend correlation: **r = −0.22**





Loyalty & Spending Patterns

- Spending & pay are strongly tied to loyalty accumulation
- Most customers earn moderate points; a smaller high-value group drives much of loyalty
- “Stretch spenders” (lower pay, higher spend) exist and are a strategic opportunity



Regression & Predictive Performance

- **Model:** Multiple Linear Regression with Pay and Spend as predictors
- Key metric: **Adjusted R^2 = 0.83** (model explains 83% of loyalty variation)
- Coefficients summary:
    - Pay: positive significant
    - Spend: positive significant

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                            OLS Regression Results
==============================================================================
Dep. Variable:                loyalty   R-squared:                       0.830
Model:                            OLS   Adj. R-squared:                  0.830
Method:                 Least Squares   F-statistic:                     3895.
Date:                Sat, 04 Oct 2025   Prob (F-statistic):               0.00
Time:                        18:54:53   Log-Likelihood:                -12307.
No. Observations:                1600   AIC:                         2.462e+04
Df Residuals:                    1597   BIC:                         2.464e+04
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -1700.3237     39.588    -42.950      0.000   -1777.974   -1622.674
spend         32.6439      0.510     63.947      0.000      31.643      33.645
pay           34.3346      0.574     59.838      0.000      33.209      35.460
==============================================================================
Omnibus:                        2.977   Durbin-Watson:                   2.034
Prob(Omnibus):                  0.226   Jarque-Bera (JB):                2.923
Skew:                           0.075   Prob(JB):                        0.232
Kurtosis:                       3.147   Cond. No.                         220.
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Customer Segmentation

- **Premium Players** – high income & spend; top loyalty.
- **Steady Spenders** – middle income; reliable base.
- **Budget Buyers** – low income; price-sensitive.
- **Stretch Spenders** – lower income, high spend; growth potential.
- **Value Seekers** – mid-income, moderate spend; respond to deals.

What are Customers Saying?

- **Positive feedback:** "Fun," "Great," "Family," "Love," "Good value."
- **Negative feedback:** "Quality," "Bad", "Poor", "Delivery"
- Most reviews are positive— strong brand sentiment overall.
- Opportunities: improve product quality.

Key Patterns & Business Implications

What We Learned

- Loyalty driven by spending & income.
- Five clear customer segments identified.
- Positive brand perception; service issues limit retention.

What It Means

- Loyalty is predictable — we can forecast and influence it.
- Segmentation enables targeted marketing.
- Operational fixes will raise satisfaction and sales.

Recommendations

- **Tiered Loyalty Programme** – Bronze/Silver/Gold to reward and retain top customers.
- **Targeted Marketing** – premium offers for high-value segments; affordable bundles for value seekers.
- **Operational Improvements** – fix product quality, stock and delivery issues highlighted in reviews.
- **Predictive Planning** – use loyalty models in CRM to forecast and personalise campaigns.
- **Data Enrichment** – add region and frequency metrics to improve future accuracy.

Future Steps & Conclusion

- Extend the dataset with regional and channel information.
- Test advanced models (Random Forest) for even better forecasting.
- Continue real-time sentiment tracking.
- **Outcome:** Loyalty is predictable and actionable — Turtle Games can now grow through data-driven decisions.